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Open Access Support provided by:
Autonomous University of Madrid

Published: 22 September 2021

[Citation in BibTeX format](#)

Interacción '21: XXI International
Conference on Human Computer
Interaction
September 22 - 24, 2021
Málaga, Spain

Heuristic-Based Usability Evaluation Support: A Systematic Literature Review and Comparative Study

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ABSTRACT

Heuristic evaluation is a popular inspection method principally used for the assessment of usability according to expert-based criteria. Since its (practical) emergence in the early 90s, it has become one of the most widely-used method, mainly utilized for formative usability assessments at large. In this paper, an overview of the different approaches for heuristic-based usability evaluation is presented. To this end, a Systematic Literature Review has been carried out, together with a comparative study to analyze different existing tools for supporting heuristic evaluation. In this way, different research questions have been proposed, which have been answered by means of the evidences found, providing an extensive analysis of different typologies, classifications, outcomes and tools related to heuristic evaluation. In general, the study concludes that there is a lack of supporting tools with advanced software architectures, featuring client/server and database facilities for further analytics that encourage comparison and benchmarking of heuristic evaluations, lacking also advanced statistical processing for helping in decision-making. According to that, this study represents a starting point for the construction of a complete supporting tool intended to provide support for usability evaluators and usability engineers overall to carry out heuristic evaluation.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI) → HCI design and evaluation methods → **Heuristic evaluations** • General and reference → Document types → **Surveys and overviews**

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Interacción '21, September 22-24, 2021, Málaga, Spain
© 2021 Association for Computing Machinery.
ACM ISBN 978-1-4503-7597-9/21/09...\$15.00
<https://doi.org/10.1145/3471391.3471395>

KEYWORDS

Evaluation Tool, Heuristic Evaluation, Usability Inspection Method, User-Centered Design, Systematic Literature Review

ACM Reference format:

Jon Fernández and José A. Macías. 2021. Heuristic-Based Usability Evaluation Support: A Systematic Literature Reviews and Comparative Study. In *XXI International Conference on Human Computer Interaction (Interacción '21)*, September 22–24, 2021, Málaga, Spain, 9 pages. <https://doi.org/10.1145/3471391.3471395>.

1 Introduction

Heuristic evaluation is a well-known usability inspection method [9, 40, 59, 64] intended to evaluate a set of previously-defined expert-based requirements called heuristics. Heuristics help find, from an expert point of view, usability problems that the user may undergo when interacting with an (interactive) artifact.

In most cases, heuristic evaluation is manually carried out, [33, 35] using custom spreadsheets and other handcraft resources. There is, in general, a certain need for specific usability tools to support the whole evaluation process [5, 34, 36], as it also happens in heuristic evaluation [17].

In this regard, the main goal of this paper is to place value on the necessity of a systematic support for heuristic evaluation, unveiling the lack of specific approaches for an effective decision-making in complex evaluations requiring precise metrics for the comparison of different assessments in a systematic way. The idea is to provide the evaluator with useful feedback based on statistical evidence. To carry out this task, this paper contributes with a Systematic Literature Review (SLR) aimed at gathering information and providing a discussion among different approaches, thus describing the methodological principles for the construction of proprietary heuristics, the methods for analyzing evaluation results, as well as the existing facilities for comparing assessments. Furthermore, a comparative study of different supporting tools is also contributed, thus analyzing main strengths and weaknesses. The main aim is to facilitate a knowledge corpus for the development and improvement of future supporting tools based on heuristic evaluation.

This paper is structured as follows. Section 2 describes the SLR method and the proposed research questions. Section 3 describes the results obtained to answer each research question. Section 4 describes a comparative study of existing supporting tools. Section 5 discusses the threats to validity related to the study carried out. Finally, Section 6 reports on conclusions and future work.

2 Systematic Literature Review

A SLR allows researchers to collect and analyze data using repeatable and analytical methods [29]. This can be applied when an accurate analysis is needed, being a suitable way to find and examine existing evidences.

2.1 Method

In general, the methodology applied may vary according to the discipline. In this paper, the method used is based on a lighter SLR version [29], also featuring a more specific approach in which the original PICOC (Population, Intervention, Comparison, Outcomes and Context) classification is explicitly adapted to the field [20].

From a general point of view, the SLR method consists of proposing a set of research questions that will drive the research and will be answered by searching related articles in scientific libraries by using a developed search string. Once located, the articles have to be screened according to inclusion and exclusion criteria, in order to obtain a set of primary studies that will be analyzed to answer each research question.

2.2 Research Questions

Based on the aforementioned argumentations, the following research questions to drive the SLR are proposed:

- RQ1: What are the most relevant approaches on heuristics, mainly related to usability and user experience (UX) evaluation?
- RQ2: What methods are used for the construction of (proprietary) heuristics in the context of usability and UX evaluation?
- RQ3: What statistical methods are the most commonly used for analyzing and comparing the results of different usability and UX heuristic evaluations?

2.3 Search String

To find out evidences and provide answers to the above research questions, a comprehensive search string has been composed. This was utilized in different search engines related to well-known digital libraries:

- (1) IEEE Xplore
- (2) ACM Digital Library
- (3) Scopus
- (4) Springer

(5) Google Scholar

Table 1 shows the proposed search string. As previously mentioned, PICOC components have been used to better define and restrict the research question. In this case, a lighter version has been utilized [20], splitting up the search string into experimental situation and environment, treatments and response variables. As shown in Table 1, the search string is specifically related to the main research questions stated, as the objective pursued is to find articles related to heuristic evaluation in the context of usability and UX, but also reporting information related to tools or supporting facilities.

Table 1: Search string split up into components

Component	String
Experimental situation and environment	("heuristic" OR "heuristic evaluation" OR "checklist")
Treatments	AND ("usability" OR user experience" OR "UX" OR "user centered design" OR "UCD")
Response variables	AND ("method" OR "support" OR "tool" OR "case" OR "statistic" OR "analysis" OR "automatic")

2.4 Inclusion and Exclusion Criteria

Inclusion and exclusion criteria were defined in order to limit the number of articles retrieved and select the most related ones.

On the one hand, the inclusion criteria applied were the following: articles written in English or Spanish in the last 11 years (from 2011 to 2021), published in journals or conference proceedings, and presenting innovative methods, techniques or works on heuristic evaluation in the context of usability and UX. Title and abstract were inspected in order to select the most suitable articles related to the target topics.

On the other hand, the exclusion criteria considered were articles not related to the target topics, short or inconclusive studies or duplicated articles.

2.5 Screening and Selection of Papers

Once the search string was applied in the aforementioned search engines, a huge and preliminary amount of articles were obtained, as can be shown in Table 2. This is very usual and happens when a first search is applied in most search engines, as they return more articles than expected/related and the set has to be refined.

Table 2: Search results

Search Engine	Total of Articles Found
(1) IEEE Xplore	401
(2) ACM Digital Library	8,368

(3) Scopus	3,222
(4) Springer	15,789
(5) Google Scholar	942,000

Therefore, a screening process was carried out to obtain a more reduced and meaningful set of articles by applying the aforementioned inclusion and exclusion criteria. Some of them were automatically applied using the facilities provided by some of the search engines, whereas in other cases such criteria had to be manually applied to filter the articles and obtain a smaller set. This reduction does not have to necessarily affect the quality of the study, as will be discussed in Section 5.

Table 3 shows the final number of articles selected, including the publication year and the related research question with the aim of answering each one with the corresponding references. As shown in Table 3, the number of articles of interest was reduced to 16. Those can be considered as the primary studies resulting from the SLR. In addition, and considering that only recent articles were initially selected, a later process of Snowball Sampling was applied by examining the references appearing in the primary studies. The aim was to find out more related bibliography.

Table 3: Primary studies related to each research question

Reference	Year	Research Question
[3]	2013	RQ3
[12]	2017	RQ2
[13]	2013	RQ2
[15]	2015	RQ2
[19]	2011	RQ2
[26]	2014	RQ2
[27]	2012	RQ2
[30]	2013	RQ2
[42]	2013	RQ1
[56]	2017	RQ2
[57]	2019	RQ2
[58]	2018	RQ2
[62]	2011	RQ2
[65]	2016	RQ2
[71]	2011	RQ3
[74]	2014	RQ2

Finally, another 32 relevant articles were found, obtaining a total of 48 different research papers. This conforms an extended set of

primary studies that will be used to answer the proposed research question with enriched argumentations.

Figure 1 shows a bubble plot with the final number of papers utilized in the study. In this plot the x axis represents the time period, whereas the y axis represents the related research question. It is worth noting that the number of papers appearing in Figure 1 is 48, whereas the total number of citations in Section 3 (to answer the research questions) is 51. This is due to the fact that one paper was utilized to answer more than one research question. More specifically, the reference related to Nielsen's heuristics (1994) [46] is first mentioned in RQ1, and later utilized in RQ2 and RQ3. Also, there is one reference in RQ1 that mentions the emergence of a specific web language, which is only used for contextualization purposes and has not been consequently included in the bubble plot.

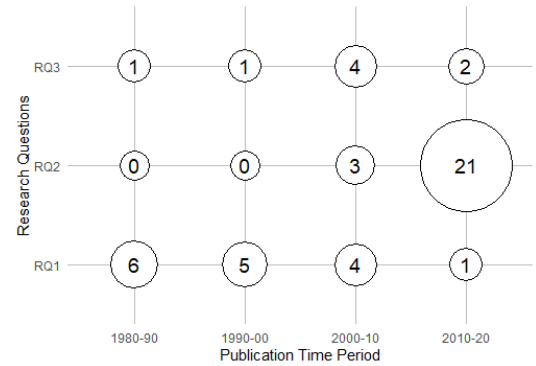


Figure 1: Papers per publication date and research question

3 Results and Discussion

The extended set of primary studies was carefully examined and analyzed. In this way, the following subsections include the main findings that are comprehensively discussed in order to answer each research question in detail.

3.1 (RQ1) What are the most relevant approaches on heuristics, mainly related to usability and UX evaluation?

There has been a great deal of research over time concerning heuristic evaluation [42].

In 1986, Smith and Mosier were among the forerunners of the heuristic principles, since they made the decision to write a set of guidelines to define the interaction between humans and machines [45]. One year later, Marshall also presented a set of heuristic principles to support human-computer interaction [41]. In the same year, Shneiderman proposed another set of heuristics called "The eight golden rules for interface design" [67]. A year later, Marlin Brown released the book called "Human Computer Interface Design Guidelines" [6], where some usability principles in the context of interactive interfaces were defined. Norman, as

Brown did, proposed in 1989 seven heuristic principles to take into account in user centered processes, appearing in the book called “The Design of Every Day Things” [50]. One year later, Molich and Nielsen proposed a new set of heuristics consisting of 9 rules to improve the human-computer dialogue [44]. In 1992, Mayhew described another set of heuristics, where the purpose was to define the design of user-centered systems [43].

There was an important turning point related to heuristic evaluation in 1994, when Nielsen proposed an entire heuristic categorization by defining a list of guidelines to cover a great variety of aspects related with the usability of interactive systems [46]. In 1995, Constantine proposed a new set of heuristics composed of five general principles and six specific ones for the design of interactive systems [10], just one year before the emergence of the PHP language that shook up the development of web applications [66]. In 1996, Gerhard-Powals also developed a set of heuristics similar to the one developed by Nielsen, but considering a more holistic approach [16].

Since its emergence, web user interfaces have become the point of focus for most heuristic-based innovations. In fact, in 1997 Instone presented a set of guidelines for website applications [25] and, in 2002, Nielsen and Tahir released the book called “Homepage usability: 50 websites deconstructed”, where a great variety of heuristic principles were described in order to improve the design of web applications [48]. One year later, Tognazzini created a webpage called “Ask Tog”, where user-interface and web-oriented principles were defined [4]. In 2004, Deniese Pierotti, based on the principles proposed by Molich and Nielsen, extended the set of heuristics by adding three more categories, thus increasing the heuristic checklist [55].

Since then, other relevant works, like the one released in 2008 by González, attempted to gather the existing heuristics principles proposed by Nielsen, Schneiderman, Instone, Tognazzini, Constantine and Mayhew [18]. The final set of heuristics were aimed at obtaining a longer set of usability principles including those from the aforementioned authors. Later on studies attempted to propose multiple variations of these typologies of heuristics, which are achieved by modifying the previous approaches defined by Nielsen [46], and adapting them to a given context/problem.

3.2 (RQ2) What methods are used for the construction of (proprietary) heuristics in the context of usability and UX evaluation?

Paddison and Englefield proposed two methods for developing heuristics [51]: (1) exploring the literature and (2) analyzing data from previous studies. However, the authors did not clarify the meaning of “analyzing data from previous studies”, which could be interpreted as conducting primary research or performing a meta-analysis of other researchers’ results [68].

Also, Ling and Salvendy suggested three methods for establishing heuristics [31]: (1) reviewing and highlighting previous research,

(2) modifying or adapting existing heuristics (i.e. Nielsen heuristics) and (3) basing the heuristics on empirical data.

On the other hand, Jaferian et. al. claimed that two approaches can be used to develop specific heuristics for a particular system [26]: (1) a bottom-up approach where qualitative data collection and analysis are used to understand domain characteristics and then create the heuristics using real-world data, and (2) a top-down approach where existing theories are used by experts to create the new set of heuristics. However, the heuristic derivation process is prone to researcher bias. To avoid such limitation, a more rigorous process can be used by combining the two approaches to create heuristics. The process can start by adopting a bottom-up approach to collect empirical data by using qualitative data analysis, and then a top-down approach can be used to justify, support, and combine the identified heuristics into a new set.

Additionally, other approaches are based on using a survey to be filled in by researchers and therefore formalize a methodology to establish specific usability heuristics [27]. In this sense, Christian Rusu et. al. proposed a set of six stages as a methodological approach for developing specific usability heuristics that could be used to evaluate particular software systems [62]. The six stages can be defined as: (1) exploratory, where information on specific topics is gathered, including general or topic-related characteristics; (2) descriptive, where the most relevant characteristics of the previously collected information is highlighted in order to formalize the main concepts associated with the topic of the study; (3) correlational, where the identification of the characteristics of each heuristic for specific applications is carried out by considering traditional heuristics [46]; (4) formally specifying the set of heuristics, making use of a standard template [27]; (5) experimental validation, where comparison between the new set of heuristics and the traditional one is carried out [46]; and (6) refinement of the heuristics, based on the feedback obtained through the experimental validation stage. This has been used in different studies for the creation of new sets of heuristics in domain-specific applications [2, 7, 65, 69, 73, 74, 13–15, 23, 52, 60–62]. However, some specific improvements are needed, such as defining iteration, identifying what activities should be performed, and specifying when it is appropriate to stop refining the set of heuristics in the last phase of the methodology [58].

Greunen et. al. proposed a three-phase process aimed at creating heuristics for specific application domains [19]. In the first phase, high-level designed heuristics are created. In the second phase, experts are provided with a validation tool to verify the heuristics. Finally, in the third phase, the high-level heuristics are applied in a specific context. A formal process for creating heuristics is suggested, clearly explaining each stage and the activities involved. Nevertheless, some deficiencies were found in this approach [56], such as: (1) it is not explained how to define or specify heuristics, (2) the authors suggest the utilization of a tool for validating the heuristics, but they do not mention what tool is or how to use it, (3) the validation with real case studies is oriented only to web sites, and (4) the experience in heuristic evaluations

and prior knowledge of usability heuristics are not listed in the selection of users at the validation phase.

Furthermore, Lechner's group proposed a methodology for developing a set of domain-specific usability heuristics [30]. The methodology comprises two phases with several activities. In the first phase, existing heuristics are rated, rewarded and new heuristics are developed. In the second phase, developed heuristics are rated, again developed, and then validated. In this proposal, the authors suggest to involve users in the validation phase.

Closely related, Quiñones and Rusu proposed a list of minimum activities to be performed when creating a new set of heuristics [56]. Those are: (1) determine the specific characteristics of the application to be evaluated in terms of the new set of heuristics, (2) identify existing sets of usability heuristics to determine how those can help define the new heuristics, (3) specify the new set of heuristics following a standard template to obtain a well-defined and easy-to-understand set of heuristics, and (4) validate the new set of heuristics to determine whether the heuristics allow to identify and detect usability problems related to the application. Later on, the authors released an extension and improvements [57] of the heuristics proposed in [62], proposing eight phases, and changing the order stated: (1) exploratory, (2) experimental, (3) descriptive, (4) correlational, (5) selection, (6) specification, (7) validation and (8) refinement.

3.3 (RQ3) What statistical methods are the most commonly used for analyzing and comparing the results of different usability and UX heuristic evaluations?

The method defined by Nielsen [46] to rate heuristics specifically considers several factors related to the severity of the problem. Those are: (1) the frequency of the problem occurrence, (2) the impact that it has on human-computer interaction, and (3) the persistence of the problem. To this end, the severity is rated in a range from 0 (not a usability problem) to 4 (catastrophe, it must be fixed). Therefore, for each heuristic, an average of this score can be obtained. It is also possible to average the scores obtained by different evaluators on the same heuristic. However, it might not make sense to average the scores of all heuristics, since it is possible that they have different weights or meanings.

The UsabAIPO initiative released the GestorHeuristicas tool, featuring statistical processing from the severity scores of a heuristic evaluation [47]. In the work proposed by González et. al. [17], a heuristic evaluation was performed using a set of 25 heuristics grouped into 4 categories (design, navigation, content and search). Each of the categories has a related weight to rate the severity score [47], obtaining a function to calculate the result of the heuristic evaluation and thus having an estimation of the level of usability of the whole system, allowing comparisons among different evaluations. However, this value does not provide concluding information about the differences, as the final percentage result is only useful in qualitative comparisons among

the evaluations, so further indicators are needed to have a more accurate benchmarking. In addition, other statistics are utilized, such as univariate analysis considering the distribution of the data, the presence of normality, symmetry, the presence of outliers, and a multivariate analysis to examine the relationship among the results belonging to different categories. Also, the degree of independence of the categories is tested using a Chi-square test. However, all these statistics are not directly calculated by the tool, and the Minitab software is needed to compute all of them.

Prometheus tool was developed by the W3C to support the heuristic evaluation performed by Sirius [71]. Unlike the UsabAIPO initiative, in Prometheus the heuristics are not grouped, and a correction factor is applied to each of the evaluation criteria to obtain different levels of usability depending on the importance of these values in the system to be evaluated.

In addition, Saaty proposed an analytical process based on decision-making for marketing purposes [63]. Closely related, Delice and Güngör utilized this method to introduce heuristic evaluation [11] using matrix operations. The approach consists of three main steps: (1) hierarchy construction, (2) priority analysis, and (3) consistency checking [22]. First, the hierarchy is constructed, so the overall objective is at the top level, criteria are at the middle levels, and alternatives are at the bottom [1]. In the second step, the relative priorities of the items at each level of the hierarchy, with respect to the items at the next higher level, are determined using a pairwise comparison matrix that is quantified using the Saaty's 1-9 scale [63]. This method helps evaluators perform pairwise comparisons among usability problems. As a result of these comparisons, it is possible to obtain a sort of severity ratings related to usability problems.

AlRoobaea et. al. presented an approach for the comparison of evaluation results [3]. In order to determine whether the proposed system has produced a sufficiently high-quality evaluation method, the results were compared using the following procedure: (1) comparison of the average time spent by each group when using each method during the evaluation sessions, (2) comparison of the results of usability problems and their severity level, in order to evaluate the performance of each method in terms of identifying unique and overlapping problems, thus identifying real usability problems, and (3) utilization of a set of statistical analyses in SPSS, such as one-way ANOVA and Pearson's correlation, in order to find out if there are differences, in each metric used, with the aim of finding usability problems. In order to make further comparisons among the performance of these methods in identifying usability problems, a set of metrics were used to examine their performance.

4 Comparative Study of Supporting Tools

In addition to the SLR accomplished, a comparative study of existing supporting tools for heuristic evaluation has been carried out. To take up this task, a comprehensive search using Google

engine was carried out, selecting the tools that appears more frequently. Also, some other tools have been directly extracted from the articles found by means of the SLR, as some of the tools are related to research and academic works. All the tools selected are within the context of heuristics for usability and UX evaluation.

4.1 Supporting Tools for Heuristic Evaluation

In general, most heuristic evaluations are carried out using spreadsheets, which is the principal supporting tool found in the search carried out. In this way, the checklist is disposed in an Excel-style spreadsheet, in order for participants to fill in the corresponding cells to indicate the rate for each heuristic. Some automatic statistical calculations are utilized, in some cases, to obtain the average and standard deviation as a final outcome. The more frequently checklists found in this case are those related to Deniese Pierotti and Jakob Nielsen [46], later on modified by Pierotti [55].

In some other cases, specific software tools are used, mostly coming from the academic context. This is the case for Prometheus tool, which features a web interface to carry out heuristic evaluations, enabling the evaluation and comparison of different websites [71]. The evaluation process is carried out based on a review by different authors. The proposed heuristics are defined in [4, 11, 18, 25, 32, 44, 53, 55, 67].

Closely related, a supporting web tool for heuristic evaluation is proposed in [24], featuring comparisons among different heuristics [16, 46]. This approach includes a comparison between paper-and-pen-based heuristics and the evaluation mechanism supported by the tool.

In addition, an Android tool for the automatic detection of usability problems has been also proposed in [54]. The tool is based on heuristics for Android mobile applications, and it automates some of the tasks involved in heuristic evaluations. Specifically, the set of heuristics comprises a reduction made by Thitichaimongkhon & Senivongse from the heuristics previously defined by Nielsen and Schneiderman [70].

Furthermore, Open-HEREDEUX tool [42] allows UX experts to evaluate the usability in a semi-automatic way. The tool comprises: (1) an open repository with multiple set of heuristics [4, 6, 48, 49, 55, 67, 10, 18, 25, 41, 43–46], (2) a heuristics advisor, (3) facilities for heuristic scoring, and (4) an analyzer of results.

Moreover, the DECIDE framework [32] includes a tool called R-IDE [28] supporting the evaluation process. Specifically, R-IDE has been designed to provide support for the following tasks: (1) scheduling of the evaluation, (2) development of scenarios, (3) estimating the cost of the evaluation, (4) handling ethical issues, (5) analyzing data and (6) writing the final report. R-IDE allows manual selection of the heuristics defined by Nielsen (allowing

own adaptations) for a single website, also enabling to define heuristic evaluations that will be manually performed or by using an external tool.

Furthermore, UsabAIPO initiative's tool [17] called GestorHeurística is designed to store information related to the experimentation, as well as data from the heuristic questions to be answered by each evaluator. The main tasks of the tools comprise: (1) store the basic information of the projects and the evaluators who will perform the evaluation, (2) select the suitable sub-heuristics among predefined sets [4, 25, 46, 67], (3) send the heuristic evaluation to the evaluators, (4) enter the assessment of each sub-heuristic together with comments, (5) receive the results of the heuristic evaluation, and (6) analyze the results of the evaluators and produce a report. As it has been mentioned before, statistical results are externally performed and obtained by using the Minitab software.

Another approach is UserPlus [21], which provides detailed guidelines that are integrated into the development stages by a conformance process. The guidelines are attached to cases that contain information on how previous projects achieved conformance to the guidelines. This provides developers with an immediate benefit by allowing them to leverage experience. The tool uses the set of heuristics defined by Smith and Mosier, but own heuristics can be also added.

4.2 Comparative Results at a Glance

In order to compare and analyze the selected tools, the following criteria were considered:

- 1 - Implemented functionality (main features and limitations).
- 2 - Kind of heuristic checklist supported.
- 3 - Implemented measures and measurement mechanisms.
- 4 - Feasibility of introducing own heuristics and measures.
- 5 - Software architecture (server / client and database support) in order to facilitate distributed and further analysis.
- 6 - Statistical processing.
- 7 - Output format and visual facilities.
- 8 - Feasibility of comparing different heuristic evaluations in order to enable benchmarking.

Regarding issue 1, principal features and limitations were analyzed in the previous subsection. In general, and except for the case of manual spreadsheet-based approaches, most of the supporting tools provide facilities to address the different steps of the heuristic evaluation, but being the processing and comparisons of results the most challenging aspects. As for issue 2, most tools are based on Nielsen, Pierotti, and Schneiderman heuristics at large, whereas others feature different or proprietary checklists. In this sense, Open-HEREDEUX and Prometheus have a large dataset of predefined heuristics. With respect to issue 3, principal measures include Likert-scale rating, average percentage values that can be also obtained from weighted formulas (as in

GestorHeurística) and other supporting metrics, as it is the case for Prometheus and Saaty approaches.

As for issues 4, 5, 6, 7 and 8, they are summarized in Table 4. In a nutshell, most analyzed tools enable the possibility of adding predefined heuristics, except for the case of R-IDE, Android tool and GestorHeurística, which does not further detail such facility. This is carried through either a simple textbox or by using a more complex mechanism where it is possible to categorize the heuristics to be introduced. On the other hand, out of the nine tools analyzed, only four had a client and server software architecture. As for statistical processing, only the severity score defined by Nielsen, usability percentages, and available formula facilities in Excel spreadsheets are used. In addition, most of the tools need external software to carry out calculations. Regarding outcomes, apart from the default options available in Excel spreadsheets, Prometheus offers outputs and comparisons among applications based on a usability percentage, and Open-HEREDEUX allows to evaluate the results through Pearson's correlation among evaluators. Only Prometheus allows comparison among heuristic results in a quantitative way, thus it is not possible to know in which aspects they differ in a qualitative way.

Table 2: Comparative Study of Supporting Tools for Heuristic Evaluation. Issues 4 to 8

Analyzed Tool	4	5	6	7	8
Nielsen and Pierotti (Spreadsheet)	Y	N	Nielsen's severity rating	Y	N
Prometheus	Y	Y	Usability percentage	Y	Y
Web tool	Y	N/A	N	N	N
Android tool	N	N	N	N	N
Open-HEREDEUX	Y	Y	Severity rating	Y	N
R-IDE	N	Y	N	N	N
GestorHeurística	N/A	N/A	N	N	N
UserPlus	Y	Y	N	N	N

5 Threats to Validity

With respect to threats to validity related to the study presented, publication and selection bias, together with search string arbitrariness, are the principal internal threats to consider, whereas generalizability can be meant as the external threat to consider in this context.

On the one hand, publication bias is related to the fact that positive results are more likely to be published than negative ones,

and thus be selected in the study. To leverage this threat, the bibliographic search has been carried out using a wide selection of proprietary libraries, but also using other search engines such as Google and Google Scholar, including also gray literature in order to find out the most relevant heuristic-evaluation supporting tools. Selection bias, referred to the distortion produced in selecting literature under wrong or insufficient criteria, has been also leveraged by widely defining inclusion and exclusion criteria related to the target domain. As for the search string arbitrariness, it has been reduced by involving two reviewers, so the search string was refined according to the goal desired.

With respect to generalizability, admittedly a reduced number of primary studies was obtained, but this has been leveraged by using the Snowball Sampling technique in order to increase the number of papers included in the analysis, which are representative for the study carried out.

6 Conclusion

Usability evaluation is a principal concern in interactive software development today [8, 34, 36–39]. Expert-based evaluation is an important part of the process, being the heuristic evaluation one of the most utilized inspection methods to evaluate an interactive system or artifact against a requirements checklist [9].

This paper is an attempt to analyze different approaches related to heuristics within the context of usability and UX evaluation. To this end, a comprehensive study, including a SLR and a comparison on supporting tools, have been carried out.

In general, the findings presented in this paper suggest that there is a considerable variety of heuristics available that can be adapted to almost every existing interactive system. Also, in recent years, new methods have been proposed for building up own heuristics for specific cases. However, no automatic methods to define own heuristics have been found, implying that heuristic evaluation still needs from human intervention for a successful completion. In addition, evaluation strategies also differ according to each approach, being specific scores (featuring mean and standard deviation), percentage weighted formulas for criteria grouping and severity levels the most commonly found. As for the feasibility of comparing results, the percentage value is the most used mechanism, featuring a manual process in most cases. Only one of the analyzed tools that comes from an academic environment provides facilities for storing and analyzing heuristics, featuring also a heuristics assessor depending on the system to be evaluated, and an analyzer for the results obtained [42], lacking however the facility of deep processing and comparison among different evaluations for benchmarking purposes.

All in all, it can be concluded that there is a lack of supporting tools with advanced software architectures, featuring client/server and database facilities for further analytics that

encourage comparison and benchmarking, lacking also advanced statistical processing for helping in decision-making.

As for future work, it is expected to use the core knowledge generated in this study to develop a supporting tool covering and improving all the drawbacks reported. The tool would be specifically focused on the comparative analysis of heuristic-based usability assessments, -i.e., usability perception benchmarking [72] based on expert criteria.

ACKNOWLEDGMENTS

This work was partially supported by the Spanish Government (project RTI2018-095255-B-I00) and the Madrid Research Council (project P2018/TCS-4314).

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